Q3) $\int \frac{dx}{(x^2-4)^{\frac{3}{2}}}$

sol

$x = 2\sec\theta$

$dx = 2\sec\theta \tan\theta \cdot d\theta$

$$\int \frac{2\sec\theta \tan\theta \, d\theta}{(4\sec^2\theta - 4)^{\frac{3}{2}}}$$

$$= \int \frac{2\sec\theta \tan\theta \, d\theta}{(4)^{\frac{3}{2}}(\sec^2\theta - 1)^{\frac{3}{2}}}$$

note. $4^{\frac{3}{2}} = (4^{\frac{1}{2}})^3$

$= 2^3 = 8$

$$= \int \frac{\cancel{2} \sec\theta \tan\theta \, d\theta}{\cancel{8}(\tan^2\theta)^{\frac{3}{2}}}$$ → Note.

$\sec^2\theta - 1 = \tan^2\theta$ ✓

$$= \int \frac{\sec\theta \cancel{\tan\theta} \, d\theta}{\cancel{\tan^3\theta}}$$

$$= \int \frac{\sec\theta \, d\theta}{\tan^2\theta}$$

$$= \int \frac{\cancel{\sec\theta} \, d\theta}{\sin^2\theta \, \cancel{\sec^2\theta}}$$

$$= \int \frac{1}{\sin^2\theta \cdot \sec\theta} d\theta$$

$\tan^2\theta = \frac{\sin\theta}{\cos\theta}$ and

$\sin\theta \times \frac{1}{\cos\theta}$

$\sin\theta \sec\theta$

$$\int \frac{\cos\theta}{\sin\theta^2} d\theta. \quad \frac{1}{\sec\theta}$$

$$\int \frac{\cos\theta}{\sin\theta \cdot \sin\theta} d\theta$$

$$\int \cot\theta \; \operatorname{cosec}\theta \; d\theta$$

integ

$$= -\operatorname{cosec}\theta + c$$

we want in Answer in x so

$$y^2 = x^2 - 4$$
$$y = \sqrt{x^2 - 4}$$

so

$$= -\operatorname{cosec}\theta + c = \frac{-x}{\sqrt{x^2-4}} + c.$$

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